

SUNSPOKE · CALCULATION NOTE

SunSpoke sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-02	Amish Chadha	Approved follow-ups (SSP-DEC-001 decision 10). Wet-weather brake blocks added to the kit (BOM line 15, 0.10 kg, USD 12); proposed wet target of 14 m from 20 km/h and the wet stop with those blocks calculated (assumed friction 0.25); 20 km/h cut-off revisited; added mass 6.88 kg; bike kit USD 208
0.4	2026-10-02	Amish Chadha	R4, R11 and motor assumption text follow Amish's decisions of 2026-10-02; no figures changed
0.3	2026-10-01	Amish Chadha	Constructable design (SSP-DDR-003). Mass, cradle retention (three band clamps over V-saddles, real band path), end stop and gate loads, fit check, cost against the value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Motor thermistor derate added to section 4; costs updated for the thermistor motor and controller
0.1	2026-09-25	Amish Chadha	

First TRL 3 sizing note (mass, energy
and range, hill and motor heating,
solar, SwapCell interface v0.3 items,
fork and torque arms, cradle retention,
braking, cost)

SunSpoke sizing calculations

The 48 V SunSpoke kit on a SwapCell pack meets its range, mass, solar charging and cost requirements on paper: about 37 km of loaded range against 30 km, 6.88 kg added against 7 kg, about 27 km of riding per day from one 100 W panel, and a bike kit of about USD 208, USD 42 under the USD 250 value-engineering target. Version 0.3 recomputed mass, cradle retention, the fit in the main triangle and cost for the constructable design of SSP-DDR-003. No requirement is shown to be not met. Three are **at risk**: R3 (motor winding reaches about 100 °C at the top of the design climb in the base case but about 139 °C in a hotter-motor case, where the thermistor derate decided in SSP-DDR-002 would slow the bike instead of cutting assist), R4 (roadster fork slots may need filing) and R11 (dry stopping distance about 8.5 m against 9 m, a 9 % margin, and about 22 m in the wet with the standard blocks; the wet-weather blocks of version 0.5 are calculated at about 11.9 m against a proposed wet target of 14 m). Sealing, repair time, cut-off timing and SwapCell latch class V1 retention (R7, R9, R10, R13) cannot be verified until hardware exists, which is TRL 4 work and on hold by Amish's instruction.

Every number in this note is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The cost figures are read from `bom/bom.csv`. Fit checks in section 9 are printed by `python cad/src/model.py`. All values are first-principles estimates; nothing here is measured. SwapCell values follow SwapCell interface v0.3 and SWC-CAL-001.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Design load case	75 kg rider, 25 kg cargo, 22 kg roadster, kit and pack (section 2)	SSP-REQ-001
Rolling resistance, drag area, air density	Crr 0.020 on dry dirt; CdA 0.60 m ² ; 1.2 kg/m ³	Upright rider with cargo
Cruise	18 km/h, rider 70 W, 30 % allowance for hills, rough surface and stop-start	SSP-REQ-001
Motor plus controller efficiency, cruise	75 %	Small geared hub at part load
Wheel radius	0.355 m	28 in (ETRTO 635) with tyre
SwapCell pack	466 Wh at 0.2C (452 Wh with minimum cells), 46.8 V, 110 mΩ, 2.85 kg; 90 % usable window	SWC-CAL-001
Motor	48 V, 250 W geared hub, about 200 rpm no-load at 48 V, gear efficiency 90 %, winding 0.45 Ω, 15 W iron and gear loss, 40 N·m peak	Typical slow-wound 250 W geared hub; to confirm against the reference motor decided on 2026-10-02, a widely sold 250 W front geared hub such as one of Bafang's (maker's data requested; winding resistance measured at TRL 4 if not published)
Motor thermal	600 J/K stator and winding, 1.2 K/W winding to ambient while riding, 120 °C practical limit (nylon planetary gears); controller derate from 110 °C on the motor thermistor	Assumed; hot case 0.60 Ω, 450 J/K, 1.5 K/W; derate decided in SSP-DDR-002
Solar	100 W panel, 4.5 peak sun hours (4.0 low), 80 % derating, 92 % boost MPPT, 95 % cell charging, 97 % pack output	SSP-PRC-001
Torque arms	5 x 20 mm mild steel (250 MPa yield), clamp 130 mm from the axle; flats lever 5 mm on bare dropouts	
Donor dropout slot	9.53 mm (3/8 in)	Assumption until the donor survey
Cradle retention	3.5 kg receiver design pack plus 1.10 kg cradle; 8 g vibration and 25 g shock (latch class V1); three band clamps at 1.5 kN tension each over 120 degree V-saddles with 1.5 mm rubber liners, friction 0.40; centroid 77 mm above the axis of a 28.6 mm tube	SwapCell interface v0.3 item V; SSP-DDR-003
Braking	0.5 s application delay; 250 N clamp per block from a 100 N hand force; block on steel rim 0.40 dry and 0.12 wet; braking radius 0.310 m	Rod brakes on steel rims
Wet-weather blocks		No maker data yet; to be replaced by the maker's figure or a wet braking test (TRL 4)

Input	Value	Basis
	Friction on a wet steel rim 0.25 (assumed); proposed wet target 14 m from 20 km/h, 1.5 times the dry target	

■ 2. Mass (R6)

Table 2. Added mass.

Part	Mass (kg)
Motor wheel, increase over the donor front wheel	1.20
Torque arms and band clamps	0.28
Controller, rubber pad and two band clamps	0.55
Receiver cradle, three band clamps, gate and draw latch	1.10
Host adapter and charge port	0.15
Pedal-assist sensor	0.10
Handlebar control and brake sensors	0.15
Wet-weather brake blocks, four (they replace the donor blocks; counted in full to be safe)	0.10
Wiring harness and fuse	0.40
Kit	4.03
SwapCell pack (SWC-CAL-001)	2.85
Added mass	6.88 (15.2 lb)

R6 (7 kg) is met with 0.12 kg margin (it was 0.22 kg before the wet-weather blocks were added). The design load case totals 128.88 kg, rounded to "about 130 kg" in the requirements. In v0.3 the cradle mass comes from the model's volumes (tray 0.44 kg, three V-saddles 0.10 kg, end stop 0.08 kg, slide strips 0.08 kg, gate 0.07 kg, catch bar 0.02 kg) plus the bought parts (three band clamps, hinge, draw latch, receptacle and fixings); the 3 mm aluminium tray replaces the concept's folded steel to hold the mass to 1.10 kg.

■ 3. Energy per kilometer and range (R2)

On a dry dirt road at 18 km/h the design case sees 25.2 N of rolling resistance and 9.0 N of drag, 12.36 Wh/km at the wheel with the 30 % allowance. The rider supplies 3.89 Wh/km, so the motor supplies 8.47 Wh/km and the pack about **11.3 Wh/km**. The TRL 2 figure of 12 Wh/km was rounded up and is superseded.

Table 3. Range on one SwapCell pack (419 Wh usable with nominal cells).

Case	Mass (kg)	Pack energy (Wh/km)	Range (km)
Design: 25 kg cargo, dirt, 18 km/h	128.8	11.3	37 (36 with minimum cells)
Light: no cargo, graded murrum (Crr 0.015, 20 % allowance)	103.8	5.6	75
Heavy: 80 kg cargo, rough dirt (Crr 0.025), 15 km/h	183.8	18.5	23

R2 (30 km in the design case) is met. A trader carrying 80 kg on rough roads gets about 23 km, so R2 depends on the load case chosen with users. At cruise the pack delivers about 204 W and 4.3 A, which heats it by only about 2.1 W; at the controller's 15 A limit the pack makes about 25 W, far inside SwapCell's 20 A rating.

■ 4. Hill climb and motor heating (R3)

On the 8 % grade at 8 km/h the loaded bike needs 127.6 N (grade 100.6 N, rolling 25.2 N, drag 1.8 N), or 284 W at the wheel. With the rider at 80 W the motor supplies **204 W** and **32.6 N·m**, about 81 % of the assumed 40 N·m peak, at about 15.8 A motor current. R3 is met on power and torque. On a 10 % grade the same 330 W gives about 7.8 km/h.

At this low speed the motor is only about 62 % efficient and loses about 127 W. From a cruise temperature of about 68 °C at 35 °C ambient, the winding reaches about **100 °C** at the top of the 500 m climb (225 s), 20 K below the assumed 120 °C limit; the limit would be reached after about 915 m. In the hot case (0.60 Ω winding, 450 J/K, 1.5 K/W, a smaller hub) it reaches about **139 °C**, over the limit. R3 is therefore **at risk**: the result turns on motor data that a named motor datasheet, and later a test, must supply. Amish decided on 2026-09-25 (SSP-DDR-002) that the motor carries a winding thermistor and the controller reduces current from 110 °C so the winding never passes 120 °C, instead of cutting assist.

Thermal derate. In the base case the winding reaches 110 °C only after about 694 m of the 8 % climb, so the 500 m design climb runs at full assist. In the hot case derating starts after about 224 m, that is, at the top of the design climb. On a climb that does not end, the derate holds the winding at 120 °C with about 11.1 A motor current in the base case and 8.3 A in the hot case, and the loaded bike slows to about 4.7 km/h and 3.7 km/h respectively with the rider at 80 W. The bike keeps moving at walking pace rather than stalling, which answers the safety concern, but in the hot case it would fall below the 8 km/h of R3 near the top of the climb. R3 stays **at risk** until a named motor's thermal data are known.

■ 5. Solar charging (R8) and charge-discharge mode

Table 4. Daily energy from one 100 W panel (Figure 2 of SSP-PRC-001 uses the mid case).

Stage	Mid, 4.5 h (Wh/day)	Low, 4.0 h (Wh/day)
Sun on panel	450	400
Panel output after 20 % derating	360	320
Charger output (92 %)	331	294
Stored in pack (95 %)	315	280
Pack output (97 %)	305	271
Daily loaded range	27.0 km	24.0 km
Days to recharge 10 to 100 % (421 Wh)	1.34	1.51

R8 (20 km per day and 2 days to recharge) is met in both cases. The peak charge power into the pack is about 74 W, or 1.47 A (0.15C).

Charge-discharge mode (SwapCell interface v0.3 item C). When the rider switches the bike on while it charges, the host adapter requests mode 4, so the display and lights run from the pack. With a 4 W load the net charge current is about 1.39 A, 28 % of SwapCell's 5 A standard charge limit. When the bike is off, the adapter requests mode 3 (charge only).

■ 6. SwapCell wake and host power (item W)

The cradle receptacle loops INTERLOCK to SGND through the 10 k Ω $\pm$ 1 % coding resistor that interface v0.3 requires. With the pack's 100 k Ω pull-up from 3.3 V the node sits at 0.30 V and draws 30 μ A, inside the pack's 8.0 to 12.5 k Ω acceptance window, so a sleeping pack wakes with no supply from the bike.

The handlebar power switch sits in series with the coding resistor (decided by Amish, 2026-09-25, SSP-DDR-002). Switching on closes the loop and wakes the pack; switching off opens it, and the pack opens its output within 1 ms, which also gives the rider a hard off. The host adapter is powered from PACK+ (after the pack enables legacy discharge 2 s after a valid loop) or from the charge inlet, through a diode-OR. If the adapter fails, the pack still runs the bike in legacy discharge-only mode at 15 A, which matches the controller's limit, so the rider is not stranded; charging then needs a working adapter or a hub dock. This relies on the pack moving from legacy discharge (state 5) to heartbeat discharge (mode 2) without opening the output; interface v0.3 does not state that transition explicitly, so it is flagged to the SwapCell project (SSP-DDR-001 item 18; raising it was decided in SSP-DDR-002), not changed here.

■ 7. Fork dropouts and torque arms (R4, R10)

At the assumed 40 N·m motor peak, bare dropouts would carry about 4,000 N on each slot face (20 N·m per side over a 5 mm lever). Torque arms clamped 130 mm up each blade carry about 308 N in total, 154 N per arm clamp, and the 5 x 20 mm arm sees about 60 MPa in bending at the axle, a safety factor of about 4.2 on mild steel yield. On the design climb the reaction torque is 32.6 N·m. Torque arms on both sides stay mandatory (R10).

If the donor slot is 3/8 in (9.53 mm), the 10 mm axle flats need about 0.24 mm filed from each slot face. That is small, but it is fabrication on the donor fork, which R4 forbids. Amish decided on 2026-10-02 that the fork is never filed for now: donors are chosen whose slots take 10 mm flats, and if 3/8 in slots are the norm, a motor whose axle flats fit them is looked for first (SSP-DEC-001). R4 stays **at risk** until the donor survey shows such donors are common. Fork fatigue under the heavier wheel is not calculated.

■ 8. Cradle retention, latch class V1 (item V)

SwapCell class V1 asks the vehicle receiver to hold the pack with no release under 8 g vibration and 25 g shocks, preloading it against its end stop with at least 330 N through an over-centre lever. At a 50 N hand force that needs a lever ratio of 6.6 or more. In the constructable design (SSP-DDR-003) the lever is a bought over-centre draw latch that pulls a drop-down gate against the pack's top end; the gate is hinged 2.5 mm above the tray and its pad sits a little below the latch, so the latch pulls about 314 N for 330 N on the pack. For the receiver design mass of 3.5 kg plus the 1.10 kg cradle, 8 g gives about 361 N and 25 g about 1,128 N.

The concept assumed each band clamp gripped the tube over a full turn. A cradle sitting on the tube cannot be gripped that way: in the constructable design each band passes through two slots in the tray, round a 120 degree V-saddle and under the tube, so it wraps about 150 degrees of a 28.6 mm tube and presses the saddle's V onto the rest. Together that gives a normal force on the tube of about 4.86 times the band tension, against 6.28 times in the concept's assumption. Three band clamps instead of two restore the margin: they resist axial slip up to about 8,740 N (margin 7.7 at 25 g, was 6.8) and rotation about the down tube up to about 125 N·m against an 87 N·m moment from a 25 g lateral shock, a margin of about 1.44 (was 1.40). The V-saddles lift the pack 7 mm, which is included in the 77 mm centroid height. The clamps must still be tightened to their stated torque and checked at service. Retention cannot be verified at TRL 3.

The end stop takes the pack's 25 g shock toward the bottom bracket. The pack bears on the 3 mm aluminium wall round the open-topped plug notch at about 0.30 MPa; the strips beside and below the notch, held by the bolted side flanges and foot, see about 62 MPa in bending, a safety factor of about 3.1 on 5052-H32 yield (193 MPa). These are simple cantilever estimates.

■ 9. Fit in the main triangle (R5)

The parametric model (`cad/src/model.py`) places the pack on a 721 mm down tube with its centre at 48 % of the tube length from the bottom bracket. With the V-saddles lifting it 7 mm, the pack clears the top tube, seat tube and head tube by 121 mm (123 mm in the concept). To come out, the gate is folded down flat, the pack's latch released and the pack slid 143 mm up the tube (clear of the 120 mm guides and the 18 mm plug), then lifted 45 mm and taken out to the left. The model checks that path step by step against every cradle part and the frame: nothing is in the way, and the pack could slide 165 mm before touching the frame. In the concept the lever's hinge block stood in the pack's path; that is fixed in SSP-DDR-003. Smaller frames need this check on the donor survey.

`python cad/src/model.py --check` runs 57 constructability checks (parts that must touch do, parts that must clear do, and the removal path); all pass.

■ 10. Braking (R11)

From 20 km/h the bike covers 2.78 m during a 0.5 s application delay, so stopping in 9 m needs 2.48 m/s². Both rod brakes on dry steel rims give about 2.72 m/s² and a stopping distance of about **8.5 m** (9 % margin on deceleration); from 25 km/h the dry distance is about 12.4 m. In the wet, block friction on steel rims falls to about 0.12 and the

distance rises to about 21.8 m. The loaded bike at 20 km/h carries about 1,989 J, about 2.5 times the 795 J of the unconverted bike at 13 km/h. R11 was specified dry and is **at risk** on a thin margin; the wet case is a safety concern whatever the requirement says. On 2026-10-02 Amish added a wet stopping target to R11, put wet-weather brake blocks suited to steel rims in the kit and set a no-wet-riding rule for trials until a wet braking test meets the target (SSP-DEC-001); in this version the wet case with those blocks is calculated. The value of the wet target is proposed, not decided: 14 m from 20 km/h, which is 1.5 times the dry target. Wet-weather blocks are assumed to give a friction of 0.25 on a wet steel rim (no maker data yet), which gives a deceleration of about 1.69 m/s² and a wet stop of about **11.9 m**. The wet target needs 1.38 m/s², so the margin on deceleration is about 23 %. At this friction the highest speed that meets 14 m wet is about 21.9 km/h, and the highest that would meet the dry 9 m figure wet is about 17.1 km/h. The 20 km/h default assist limit therefore stays: it meets the proposed wet target, and wet riding stays banned in trials until a wet braking test (TRL 4, on hold) shows the real friction. R11 stays **at risk** because the wet friction is an assumption.

■ 11. Cost (R12)

Value-engineering target: USD 250 for the bike kit (`budget_usd` , a hypothetical control target, not a limit).

Estimated cost of the constructable design: USD 208 for the bike kit (USD 42 under the target).

Table 5. Cost from `bom/bom.csv` (indicative prices).

Group	Items	Cost (USD)
Bike conversion kit	1 to 5, 7 to 9, 14, 15	208
Solar charging set, costed separately	10 to 13	90
Kit plus solar set, pack excluded		298
SwapCell pack, priced once in SwapCell (SWC-CAL-001)	6	414
Full system for one rider with own panel	all	712

R12 (bike kit USD 250 or less; solar set costed separately; pack excluded) is met, USD 42 under the value-engineering target. Generic parts make up about 84 % of the kit plus solar set cost (R9 asks for 70 %). Making the design constructable (SSP-DDR-003) moved the cradle from USD 28 to USD 33 (aluminium tray, V-saddles, a third band clamp, hinge and draw latch), hardware from USD 8 to USD 11 (four more band clamps and the controller pad) and the panel stand from USD 10 to USD 14 (timber, coach bolts); the bike kit rose from USD 188 to USD 196 and the solar set from USD 86 to USD 90. Adding two pairs of wet-weather brake blocks (line 15, USD 6 a pair, an indicative price and not a quote) raised the bike kit from USD 196 to USD 208.

■ 12. Results against requirements

Table 6. Requirement status. Not met: none. Items at risk and not verifiable are listed first.

ID	Value (SSP-CAL-001)	Target	Status
R3	100 °C winding at the top of 500 m from 35 °C (139 °C in the hot case); 32.6 N·m; 204 W motor; hot case derates after 224 m	8 % for 500 m at 8 km/h at 35 °C; thermistor derate from 110 °C, no cutout	At risk
R4	Slot filing about 0.24 mm per side on a 9.53 mm slot; fitting time not calculable	No fabrication; 90 min or less	At risk
R11	8.5 m dry (9 % margin); 21.8 m wet with standard blocks, 11.9 m wet with wet-weather blocks (assumed friction 0.25)	9 m or less from 20 km/h, dry dirt; wet target added 2026-10-02, proposed value 14 m	At risk
R7	IP ratings by part selection only	IP65 electronics, IP54 motor, 150 mm water, 0 to 45 °C	Not verifiable at TRL 3
R9	Generic parts 84 % of kit cost; swap time not calculable	Pluggable joints, 20 min swap, 70 % generic	Not verifiable at TRL 3
R10	Arm clamp 154 N each, arm safety factor 4.2; 0.5 s cut-off needs a bench test	Brake cut-off, 0.5 s stop, fuse, torque arms	Not verifiable at TRL 3
R13	Coding node 0.30 V; mode 4 net 1.39 A; clamp rotation margin 1.44 at 25 g	SwapCell interface v0.3 items W, C and V1	Not verifiable at TRL 3
R1	250 W rated motor, 20 km/h cut-off, pedal assist and 6 km/h walk assist	250 W; 25 km/h or lower; assist only when pedaling	Met (by specification)
R2	37 km (36 km with minimum cells)	30 km or more	Met
R5	V-saddles for 28 to 32 mm tubes; motor for 100 mm spacing and 635 rim; pack fits and comes out of the main triangle in the model	Common roadster	Met (on paper)
R6	6.88 kg	7 kg or less	Met
R8	27 km/day; recharge 1.3 days (1.5 at 4 h)	20 km/day; 2 days or less	Met
R12	Bike kit USD 208 (USD 42 under the value-engineering target); solar set USD 90 costed separately	Bike kit USD 250 or less (value-engineering target)	Met

■ 13. Checks against earlier documents

The TRL 2 figures in SSP-PRC-001 v0.2 were checked against this script and corrected in v0.3: pack energy 12 to 11.3 Wh/km, design range 35 to 37 km, light-case range 60 to 75 km (the light case is now defined), daily range from one panel 25 to 27 km, added mass 6.5 to 6.65 kg (pack 2.8 to 2.85 kg, host adapter and V1 lever added), pack cost \$370 to \$414 (SWC-CAL-001), kit plus solar set \$261 to \$269, wheel torque on the grade 46 to 45.3 N·m and motor torque 40 to 32.5 N·m (the TRL 2 figure used the whole wheel torque less a rounded rider share), and the pack position on the down tube 52 % to 48 % (section 9). The 36 V pack option (Option B) is no longer assessed; it remains a later low-cost variant.

Safety: These calculations concern a 468 Wh lithium-ion pack, a loaded bicycle at up to 20 km/h with weak wet brakes, and new loads on an old steel fork. They are paper estimates and do not replace inspection of each donor bike or testing. Nothing may be built or ridden from this note; building and testing are TRL 4 work and on hold.